Ref. Ares(2021)4337948 - 03/07/2021

ESSnet Big Data II

Grant Agreement Number: 847375-2018-NL-BIGDATA

https://ec.europa.eu/eurostat/cros/content/essnet-big-data-0_en

Workpackage WPC

Implementation – Enterprise Characteristics

Deliverable C1: ESS web-scraping policy template

Final version, 15 July 2019

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ESS Web scraping policy template

Preamble

The purpose of this policy is to ensure that web scraping activities of European Statistical System (ESS) members, namely Eurostat and the national statistical institutes (NSIs) are carried out transparently, consistently, ethically, respecting the acquis communautaire as well as national legislation. Web scraped data should be extracted and used in an appropriate and ethical manner that limits burdens on website owners and survey respondents to the greatest extent possible.

For the purpose of this policy, webscraping is defined as the automated extraction of data available on the World Wide Web (www).[1] Web scraping is a legitimate data extraction method, complementing survey data collection and administrative record analysis as sources for European and national statistics compilation. As the extracted data are used only for statistical purposes, NSIs are *a priori* in accordance with the Regulation (EC) No 223/2009 of the European Parliament and of the Council on European statistics (1), European Statistics Code of Practice (2), General Data Protection Regulation (GDPR) (3) and national statistical legislation.

This document is, outlining key principles for good webscraping practice within the ESS. These principles are relevant for all ESS members carrying out web scraping. Individual NSIs may adopt more stringent rules complying with national legislation, conforming to national legal or cultural context, or when scraping sensitive or personal data.[2]

Background

The ESS and its members are committed to using new data sources to produce statistics and analyses across the EU (5). Web scraping allows for more timely collection of novel forms of data to produce statistical information such as companies' engagement in e-commerce, which can significantly supplement and enhance the information gleaned from traditional surveys, even while reducing response burdens, and increasing sample sizes.

Simply put, rather than requiring staff to answer questions about their employers, web scraping involves automated collection of data which companies make publicly about themselves and their goods and services.

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[1] For a more detailed definition of webscraping, see the Glossary in the Appendix.

[2] Although scraped data is freely available on the world wide web and is hence by definition open data, occassionally, particularly in cases of small enterprises data or self-employed persons, personal data such contact information, e-mails, phone numbers may be included.

Scope

This policy is intended to guide the ESS and member NSIin its web scraping activities. This includes the extraction collection, processing and use of web scraped data. The data might be scraped by statistical offices, be gathered by third party/intermediary providers, or provided by aggregator websites.

The ESS members will seek to ensure that best practices and procedures outlined in this policy are followed when procuring web scraping services or web scraped data.

Principles

Web scraping will be performed in adherence with the principles of the European Statistics Code of Practice, and in compliance with intellectual property legislation (national copyright laws and the Database directive) (6).

The members of the ESS should use web scraped data solely for statistical purposes as laid down in regulation (EC) No 223/2009 of the European Parliament and of the Council on European statistics and the applicable national statistical legislation.

The principles guiding web scraping activities should be to maximise the benefits while minimising any burden, risks or potential impacts arising from these activities.

To this end, members of the ESS should:

- seek to minimise burden on the website owners;
- identify themselves to the scraped website;
- protect all personal data according to the GDPR;
- abide by all applicable EU and national legislation;
- respect the website scraping policies, being in agreement with the statistical principles laid down in regulation 223/2009 on European statistics;
- scrape for the purpose of creating new statistical information from the data;
- ensure transparency concerning the tools, and the methods and processes that are used for web scraping as well as the methods used for producing the relevant statistics;
- ensure that statistical information is produced in an unbiased manner, e.g when preparing training data sets for machine learning;
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Practices (Implementation guidelines)

When web scraping, members of the ESS should:

- Respect the robots.txt exclusion protocol and only follow links to the extent necessary for maintaining the quality of statistics;
- Respect the wishes of website owners as set in terms and conditions insofar as practical to check those terms and scraping is not essential to maintaining the quality of statistics as implied by statistical law;
- Identify themselves in the user-agent string, and provide contact channels. This could include a link to a web-page explaining the scraper's purpose and what data it collects, responsible

team contact details, and information on how to opt-out and request that extracted data be deleted;

- Follow internet standard conventions for scraping, such as standards established by the W3C consortium;
- Be transparent about its web scraping activities, possibly by providing information on the associated website;
- Inform website owners if a considerable amount of data is extracted on a regular basis. This would not be the case if a website is scraped with a low frequency and not is scraped in full depth;
- Seek to minimize burden on web servers, by:
    - adding idle time between requests,
    - scraping at a time of day during which the web server is not expected to be under heavy load,
    - optimise the scraping strategy for minimising the number of requests to a domain;
- Only scrape data within the scope of the statistical office's legal mandate, and do not re-use or distribute the raw data;
- Handle web scraped data securely;
- Avoid web scraping in using public APIs or other data provision options where available.

Roles and responsibilities

NSIs should inform staff involved in web scraping of this policy and ensure compliance of staff with this policy in any web scraping related activities. Statistical offices should provide the required infrastructure to enable compliance. Where third party companies or individuals are contracted to scrape or provide scraped data, the NSI must ensure that the data was, or will be, extracted in line with the principles and practices above.

Governance

This policy template has been drafted as part of the work undertaken by the ESSnet Big Data project on the integration of big data sources into official statistics.

Appendix – Glossary

Application Programming Interface (API) - In computer programming, an application programming interface (API) is a set of subroutine definitions, communication protocols, and tools for building software. In general terms, it is a set of clearly defined methods of communication among various components. A good API makes it easier to develop a computer program by providing all the building blocks, which are then put together by the programmer.

Depth of crawl - The extent to which a web-crawler crawls pages "deep" within the website. If a website's homepage is referred to as "level 0", and pages linked to from the homepage are "level 1" and pages linked to from level 1 pages are "level 2" and so on, then limiting the depth of crawl means limiting the level that the web-crawler will penetrate to.

E-commerce - Commercial transactions conducted electronically on the internet, for example, online shopping.

Generic web scraping - when no a priori knowledge on the structure and content is available and the whole website must be scraped and processed in order to infer some information of interest.

Robots.txt exclusion protocol - A widely-used protocol that allows website owners to prevent any web scraping, to limit web scraping to search engines only, or to shield parts of their website from web scraping. For more details, see the robots.txt website.

Specific web scraping - *W*hen both the structure and the (type of) content of websites to be scraped are perfectly known. In this case, crawlers just have to replicate the behaviour of a human being visiting the website to collect the information of interest. Typical examples of specific web scraping can be found in the area of price statistics, where most of the items in a web store have the same product listing or product page and scraping software can retrieve specific elements, such as the original price, the sales price, the label, the description, the quantity, colour, material etc. from many web pages for many products

User-agent string - When a browser or web scraper accesses a webpage it provides a "user-agent string" to the server hosting the website and this string is then viewable by the website owner. It is possible, when building a web-scraper, to modify this user-agent string so that it contains custom text – for example, to identify the operator or purpose of this web-scraper.

Web scraper - The software or program used to carry out web scraping.

Web scraping - The extraction of data automatically available on the World Wide Web (www).

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